

Supplementary Material — Not for Review

Data Access Instructions and Open Science Commitments

Geodesic Diagnostic Trajectories (GDT)

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September 2026

This document contains administrative and logistics information for journal handling. It is not subject to peer review.

1 Data Availability Statement

All datasets used in this study are publicly available:

- **MIMIC-III** (v1.4): <https://physionet.org/content/mimiciii/>
- **MIMIC-IV** (v2.2): <https://physionet.org/content/mimiciv/>
- **eICU Collaborative Research Database** (v2.0): <https://physionet.org/content/eicu-crd/>
- **HiRID** (v1.1.1): <https://physionet.org/content/hirid/>

Access requires PhysioNet account registration and signing of the respective data use agreements. No new data were generated in this study.

2 Code and Reproducibility Commitment

Upon acceptance of this manuscript, the authors commit to:

1. Release all evaluation code, model weights, and configuration files on GitHub under a permissive open-source license (MIT).
2. Deposit a static archive of the released code on Zenodo, linked from the published manuscript.
3. Maintain a public issue tracker for reproducibility questions for a minimum of two years following publication.

The repository URL will be: [https://github.com/\[to-be-disclosed-post-review\]](https://github.com/[to-be-disclosed-post-review])
(Disclosed upon acceptance to preserve double-blind review integrity.)

3 Open Science Framework Pre-registration

Pre-registered analysis plan is deposited at the Open Science Framework (OSF). The OSF record includes:

- Pre-specified primary and secondary outcomes (as reported in Section IV of the manuscript)
- Pre-specified statistical analysis plan (Wilcoxon signed-rank tests, $\alpha=0.05$, Cohen's d)
- Pre-specified four-arm crossover design parameters
- Date of pre-registration: prior to data analysis

OSF URL: [https://osf.io/\[to-be-disclosed-post-review\]](https://osf.io/[to-be-disclosed-post-review])

4 Author Contributions (CRediT Taxonomy)

- **Hemanth Manchabale Papachappa:** Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization.
- **Aniriuddha Ganguly:** Conceptualization, Writing – Review & Editing, Supervision, Project Administration.

5 Ethical Statement

This study used exclusively de-identified, publicly available datasets. No new human subjects were enrolled and no new clinical data were collected. No IRB approval was required for this study design. The study complies with the Declaration of Helsinki in spirit, and with HIPAA Safe Harbor de-identification standards as implemented by the respective dataset custodians.

6 Article Processing Charge (APC) Statement

JTEHM is a fully open-access IEEE journal. The authors confirm awareness of the applicable APC and their intention to pay upon acceptance, subject to any applicable waivers or institutional agreements.